

1. Data Acquisition

Raster layers



Landsat-8



SRTM



ESRI-LULC



TerraClimate



ERA5-Land

Vector layers



Canadian Wildland Fire Information System (CWFIS) 2000-2024



Natural Resources Canada

2. Predictor Variables



Topographic

- Aspect
- Slope
- Elevation
- TWI
- Profile curvature
- Plan curvature



Environmental

- Land use / land cover
- Distance from rivers
- Distance from households
- Distance from roads



Vegetation

- NDVI (Normalized Difference Vegetation Index)

All predictors resampled to 1.5 km and harmonized to a common grid (BC Albers, EPSG:3005)

3. Spatiotemporal Analyses (2000-2024)



Fire inventory

National fire database fire point data

Annual & monthly fire trends

District-wise analysis



Burned area

National burned-area composite data

Fire density maps (annual · monthly · seasonal)

District-wise analysis



Climate

Climatic data (2000-2024)

Climate variability & trends

Temperature · Wind speed · Relative humidity · Precipitation · Evapotranspiration · Drought severity index · Soil moisture

4. Data Preparation & Sampling



Inventory maps preparation



Wildfire points 2000-2024



1,992 fire, 1,992 non-fire points



Raster to point (1.5 km grid)

Implemented in ArcGIS Pro 3.6

5. Modeling Pipeline



Dataset split (70% training/30% testing)



Preprocessing & feature selection (RF-RFE)



10-fold spatial cross-validation



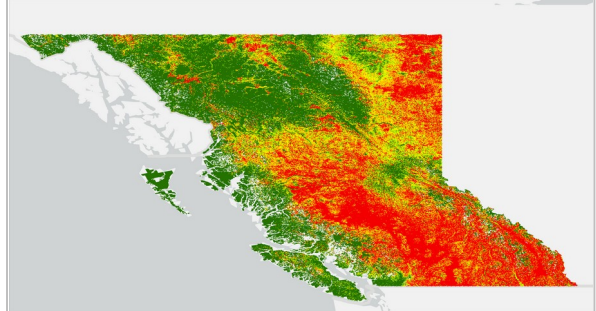
DL models (LSTM & BiLSTM)



Hyperparameter optimization (PSO)

7. Outputs & Exposure Assessment

Wildfire susceptibility mapping

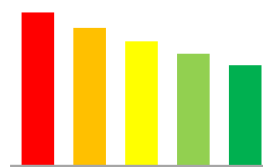


BiLSTM-PSO · AUC = 0.92 · 1.5 km resolution

Exposure assessment

Very High susceptibility zones

Top ten districts



Trees



Road network



Households



Cross-border consistency

Comparison with U.S. FSim estimates

Consistent high-susceptibility patterns across the border

6. Validation & Interpretability



Model evaluation

Accuracy · Sensitivity · Specificity · Precision · F1-score · ROC-AUC · PR-AUC



Interpretability SHAP analysis

Influential-variable assessment global & local feature attributions